

DOCUMENTATION: CUTE MAGIC - KITCHEN ASSET PACK

1. Introduction

Welcome to the **Cute Magic - Kitchen Asset Pack**! This is a massive collection of high-quality, mobile-optimized 3D assets designed to help you create stunning kitchen environments in minutes.

- **Total Assets:** ~600 unique models and prefabs.
 - **Theme:** Modern and Stylized Kitchen Interior.
 - **Optimization:** Ultra-lightweight, designed for Mobile, VR, and low-end devices.
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2. Folder Structure

The package is organized into a clear hierarchy within the `CuteMagic_Room_Kitchen` folder to ensure a smooth workflow:

- **Model:** Contains nearly 600 base 3D models in .fbx format. All models are optimized with clean topology.
 - **Prefab:** Corresponding to the models, these are ready-to-use assets. Each prefab is pre-configured with the custom shader and materials. **(Recommended for use)**.
 - **Scenes:** Includes a **Demo Scene** where all assets are arranged. You can use this scene to see how the lighting and materials interact or as a starting point for your level design.
 - **ShaderMaster:** Contains the core logic of the pack — the **ShaderMaster_02** (HLSL code). This is the custom-coded shader used across the entire pack.
 - **ShaderTexture:**
 - Contains the **Texture Atlas** (All 600 assets share one primary texture).
 - **Materials:** This sub-folder contains the material files that utilize our custom PBR shader.
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3. Technical Analysis: ShaderMaster_02

The heart of this pack is the custom shader: `CuteMagic/Mobile_HighPerformance_PBR`. Here is what makes it special:

PBR Workflow on Mobile

Unlike standard mobile shaders, this custom HLSL code brings **Physically Based Rendering (PBR)** to performance-sensitive devices. It uses a **Channel Packing** technique to minimize memory usage:

- **R (Red):** Metallic
- **G (Green):** Ambient Occlusion (AO)
- **B (Blue):** Roughness (Converted to Smoothness in-code)

Extreme Optimization

- **Draw Call Efficiency:** Since all 600+ assets share a single Material and a single Texture Atlas, Unity can use **SRP Batching** or **Static Batching** to render the entire kitchen in just a few draw calls.
 - **Universal Render Pipeline (URP):** Fully compatible with URP 2021.3 LTS and newer.
 - **Custom Sliders:** Users can adjust Metallic, Roughness, and AO intensity directly on the material without editing textures.
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4. How to Use

1. **Importing:** Ensure your project is set to **Universal Render Pipeline (URP)**.
 2. **Using Assets:** Drag and drop prefabs from the Prefab folder into your scene.
 3. **Demo Scene:** Open the scene in the Scenes folder to see the recommended lighting and layout settings.
 4. **Customization:** If you want to change the "feel" of the kitchen (e.g., make it shinier or more matte), select the material in ShaderTexture/Materials and adjust the **Metallic** or **Roughness Multiplier** in the Inspector.
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5. Support & Contact

If you encounter any issues or have questions regarding the assets, please feel free to contact:

- **Email:** cutemagic15081998@gmail.com

Thank you sincerely for choosing the Cute Magic - Kitchen Asset Pack and for being a part of my creative journey. I have poured my heart into every asset to ensure the best quality and optimization for your projects. Your support means the world to me, and I look forward to bringing you even more professional tools in our upcoming Cute Magic releases!